

6. (a) **X, Y and Z** are solutions of ethanoic acid, hydrochloric acid and sodium chloride, **but not necessarily in that order**. Reactions of **X, Y** and **Z** with magnesium gave the results shown below.

Solution	Reaction with magnesium
X	rapid fizzing, salt formed, temperature increase of 18 °C
Y	no reaction
Z	slow fizzing, salt formed, temperature increase of 11 °C

Use the table to identify **X, Y** and **Z**, giving your reasoning. Explain the results recorded and include equations to support your answer. [6 QER]

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- (b) State how adding sodium hydroxide solution allows solutions containing iron(II) and iron(III) ions to be identified. Give the observations made for both solutions. [2]

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